

SUNNE, Sweden

WMO: 024230

Lat: 59.8639N

Lon: 13.1166E

Elev: 358

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	1.9	-8.0	3.6	-3.9	-1.9	5.0	2.3	20.3	39.8	17.6	39.2	0.8	260	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	79.7	64.4	76.4	63.0	73.1	61.5	66.8	75.5	65.0	72.9	63.2	70.3	5.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.8	89.8	69.6	62.0	84.3	68.0	60.3	79.1	66.4	31.6	75.7	30.2	73.1	28.9	70.5	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.3	13.2	11.5	-9.3	83.4	7.3	3.1	-14.6	85.6	-18.8	87.5	-22.9	89.2	-28.3	91.5		
			-9.9	69.8	6.6	2.5	-14.6	71.6	-18.5	73.1	-22.2	74.5	-26.9	76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	24.3	25.5	31.6	40.8	49.5	57.1	61.3	58.9	51.4	42.0	34.3	26.9
	DBStd	14.92	10.63	9.18	8.37	5.81	6.42	4.76	4.92	4.59	5.35	6.78	7.47	10.87
	HDD50	3918	797	686	569	283	86	3	0	2	47	260	470	715
	HDD65	8404	1262	1106	1034	727	480	240	138	195	408	714	920	1180
	CDD50	1022	0	0	0	6	72	216	350	277	90	11	0	0
	CDD65	31	0	0	0	0	0	3	23	5	0	0	0	0
	CDH74	560	0	0	0	0	31	123	329	77	0	0	0	0
	CDH80	87	0	0	0	0	2	15	66	4	0	0	0	0
	WSAvg	4.6	4.5	4.4	4.5	4.6	4.8	4.7	4.3	4.2	4.5	4.6	5.0	4.6
	PrecAvg	28.5	1.8	1.4	1.3	1.5	2.7	2.8	3.3	3.3	2.6	2.8	2.6	2.0
Precipitation	PrecMax	38.4	4.6	2.6	3.1	3.7	5.3	3.9	7.8	6.1	5.3	7.3	7.0	3.5
	PrecMin	20.9	0.4	0.6	0.2	0.3	0.8	1.3	0.9	1.1	0.8	0.6	0.8	0.5
	PrecStd	4.1	1.1	0.6	0.7	0.9	1.1	0.7	1.6	1.7	1.3	1.6	1.5	0.9
	PrecStd	4.1	1.1	0.6	0.7	0.9	1.1	0.7	1.6	1.7	1.3	1.6	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	45.1	45.8	57.7	67.5	77.8	82.5	86.0	79.8	71.4	60.3	52.6	50.0	
	0.4% MCWB	42.1	40.8	47.1	52.5	59.6	64.6	65.7	64.5	60.8	54.1	49.7	47.5	
	2% DB	41.8	42.2	51.1	62.8	72.9	77.5	81.4	76.3	67.2	56.9	50.0	46.1	
	2% MCWB	39.3	39.1	43.1	50.1	57.6	62.0	65.8	64.1	58.8	52.6	47.9	44.1	
	5% DB	39.6	39.7	46.5	57.8	68.4	73.7	77.9	72.9	63.9	54.7	47.6	43.2	
	5% MCWB	37.8	37.5	40.2	47.3	55.1	60.4	64.9	62.6	57.1	51.3	45.7	41.4	
	10% DB	37.3	37.6	42.7	53.8	63.0	69.7	73.8	69.5	61.0	52.5	45.3	40.6	
	10% MCWB	35.6	35.7	37.9	45.3	52.5	58.1	62.9	61.1	55.6	49.9	43.6	39.1	
	0.4% WB	42.8	42.5	47.6	54.7	62.6	66.6	70.4	68.2	62.7	56.2	50.6	47.6	
	0.4% MCDB	44.5	44.0	56.1	65.3	72.3	76.0	78.9	74.7	67.8	58.4	51.5	49.5	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	39.9	39.7	43.8	50.9	59.4	64.2	67.7	65.7	60.2	53.9	48.1	44.3	
	2% MCDB	41.1	41.4	50.3	59.9	69.2	73.6	77.3	72.4	65.0	55.9	49.8	45.7	
	5% WB	38.0	37.9	41.1	48.4	56.7	61.8	66.0	63.9	58.3	52.0	46.0	41.7	
	5% MCDB	39.3	39.5	45.3	56.5	65.7	71.3	75.0	69.9	62.5	53.9	47.4	42.8	
	10% WB	35.8	35.7	38.9	46.0	54.0	59.7	64.2	62.3	56.6	50.2	43.7	39.1	
	10% MCDB	37.0	37.2	42.1	52.4	61.6	68.1	72.0	67.9	60.3	52.5	45.0	40.4	
Mean Daily Temperature Range	MDBR	10.1	11.8	16.2	19.7	20.0	20.1	19.4	18.7	17.3	12.5	8.8	9.8	
	5% DB	9.8	12.7	21.8	28.9	30.3	29.1	27.1	24.5	21.5	14.4	11.0	10.5	
	5% WB	9.3	9.9	14.7	17.4	16.3	15.6	13.6	13.7	13.9	10.9	10.2	9.9	
	MCDBR	9.7	11.3	19.2	25.7	23.9	25.2	23.7	20.8	18.9	12.3	10.5	10.1	
	MCWBR	9.3	9.3	13.5	16.0	13.9	14.5	13.1	12.6	13.1	10.2	10.2	9.8	
	MCWBR	9.3	9.3	13.5	16.0	13.9	14.5	13.1	12.6	13.1	10.2	10.2	9.8	
Clear-Sky Solar Irradiance	taub	0.261	0.303	0.328	0.366	0.365	0.355	0.375	0.367	0.336	0.325	0.283	0.238	
	taud	2.211	2.298	2.364	2.348	2.413	2.462	2.422	2.441	2.517	2.457	2.292	2.129	
	Ebn at Noon	168	223	256	265	274	279	269	262	253	216	159	128	
	Edh at Noon	13	21	27	33	33	32	32	30	23	18	12	9	
All-Sky Solar Radiation	RadAvg	93	285	770	1210	1544	1776	1585	1244	828	368	120	61	
	RadStd	16	56	81	157	151	109	207	128	100	71	31	7	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page